

# Unbreakable Steganography with LLMs

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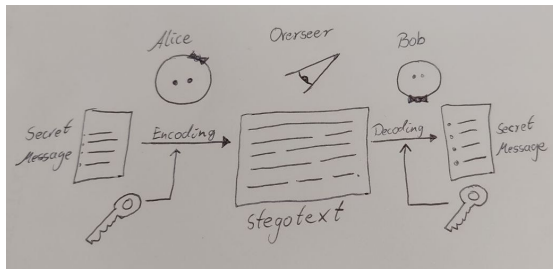
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## 1 Overview

## 2 Main Content

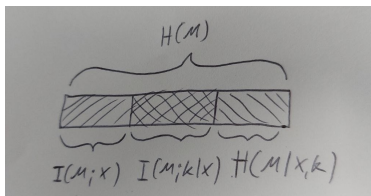
# Steganography Setup



## Ingredients:

- Sender/Encoder  $E$  and Receiver/Decoder  $D$
- Communication Channel  $C$
- A secret key  $k$  shared by  $E$  and  $D$
- A distribution  $\mathcal{D}$  of innocent messages
- A message space  $M$

# Can we have perfect perfect information theoretic security?



## Perfect Steganography:

- M: Message, X: Stego-Text, K: Key
- Perfect Security:  $I(M; X) = 0$
- Perfect Decoding:  $H(M|X, K) = 0$

$$I(M; X, K) = H(M) - H(M|X, K) \quad (\text{Def of Mut. Info}) \quad (1)$$

$$I(M; X, K) = I(M; X) + I(M; K|X) \quad (\text{Chain Rule}) \quad (2)$$

$$\Rightarrow H(M) = \underbrace{H(M|X, K)}_{=0} + \underbrace{I(M; X)}_{=0} + I(M; K|X) \leq H(K) \quad (3)$$

$$\Rightarrow H(M) \leq H(K) \quad (4)$$

- Basic Idea: Security against polynomially bounded adversaries
- Uses Pseudo-Random Functions (PRF), i.e., deterministic functions

$$F = \left\{ F_k : \{0, 1\}^{|x|} \rightarrow \{0, 1\}^{|y|} \right\}_{k \in \{0, 1\}^{|k|}},$$

[  $|k|, |x| = \text{poly}(|k|), |y| = \text{poly}(|k|)$  are key, input and output length]

- where, for any poly-time distinguisher  $D$ , the advantage

$$\text{Adv}_{F,D}^{\text{PRF}}(|k|) := \left| \Pr_k \left[ D^{F_k}(1^{|k|}) = 1 \right] - \Pr_f \left[ D^f(1^{|k|}) = 1 \right] \right|,$$

where

$$k \sim \mathcal{U} \left( \{0, 1\}^{|k|} \right) \quad \text{and} \quad f \sim \mathcal{U} \left( \{0, 1\}^{|x|} \rightarrow \{0, 1\}^{|y|} \right)$$

is negligible, i.e.,

$$\text{Adv}_{F,D}^{\text{PRF}}(|k|) < \frac{1}{\text{poly}(k)}$$

# PRF + LLM = Unbreakable Steganography

- Any signal that has entropy (or random degrees of freedom), can be turned into an unbreakable stego scheme via PRFs
- Basic Idea:
  - Sender and Receiver share PRF scheme and secret key  $k$
  - Sender uses LLM to generate distribution over next token
  - The PRF with the next symbol as input generates a pseudo-random sample
  - The receiver eliminates every symbol that wasn't compatible with the tokens so far until only one symbol remains
  - Sender moves on to next symbol

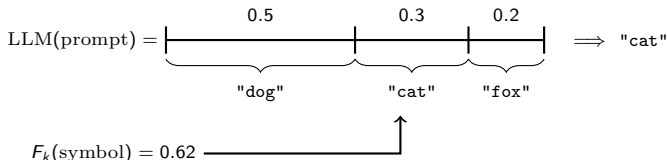


Figure: Using a PRF to select from the next-token distribution

- **Scenario 1:** LLM and prompt are known  $\rightarrow$ . Receiver can decode perfectly
- **Scenario 2:** Only tokenizer is known: Receiver can still decode probabilistically with error correction ( $\leftarrow$  symbol)
- Receiver aggregates score for every symbol
- Once score exceeds threshold  $t$ , symbol is accepted
- If a wrong symbol got accepted, the sender sends a  $\leftarrow$  correction symbol
- As long as the error rate is  $< 0.5$ , the process has net progress towards correct symbols

Questions?